

# Single-Port Product Capture and Context-Safe Weight Broadcast for Event-Driven Spiking Optical Flow

Authors to be inserted

**Abstract**—Low firing activity does not automatically accelerate event-driven optical flow: repeated arithmetic and weight delivery remain constrained by finite storage, ports, and accumulator ownership. This brief presents two exact reuse islands for a frozen spiking optical-flow workload. Both decide reuse before the resource being suppressed and preserve private completion state. C1 executes a Prosperity-compatible exact-subset forest in a single-1RW store, issues the residual, suppresses unused-parent writes, and retires completion atomically. On 51.84 million source rows, its same-ledger cycle model reduces component time by 40.99% (1.6945 $\times$ ) versus strongest-zero skipping; a mapped 28-nm nine-SRAM island occupies 166,514  $\mu\text{m}^2$  and meets 3-ns setup/hold. C2 shares one Acc24 fabric across eight typed signed sources. Token-set broadcast (TSBG) reuses only weight delivery while keeping products, destinations, and accumulators private. Across 2,880 fixed real-activity workloads spanning all 12 FC1 and 12 FC2 layer identities, same-port/cache VCS reduces post-load cycles by 45.49% (1.8345 $\times$ ) and scalar weight reads by 58.13%. A matched clock-gated implementation adds 0.50% logic area. Against equal-bandwidth K1 $\times$ 8, K8 uses 77.61% less logic and provides 4.541 $\times$  directed-throughput/logic-area. Results remain component-level and prelayout; no whole-network speedup is inferred.

**Index Terms**—Digital accelerator, event-based vision, optical flow, product sparsity, single-port SRAM, spiking neural networks.

**E**VENT cameras make inactivity visible, but sparse arithmetic is only one term in execution time. In a binary-event optical-flow deployment, nonzero sources still trigger parent-state reads, residual issues, and psum writes. A design that reports only operation sparsity can therefore be slower than an equal-resource zero-skipping baseline once the parent working set is finite and the store is single-ported.

Prosperity [1] and Phi [2] show that useful SNN speedups come from executable reuse or patterns, not firing rate alone. Digital neuromorphic processors such as ODIN [3] and event-stream accelerators such as SpinalFlow and SNE [4], [5] establish spike-driven datapaths. TCAS-II SRAM compute-in-memory macros exploit sparsity inside the array [6], [7]. Our focus is the execution circuit for repeated *products* of a convolution-like layer within a 240-KiB-class total storage budget and a physical 1RW contract, not a new definition of product sparsity.

This brief addresses two costs that remain after zero-source skipping: retaining reusable products and repeatedly delivering

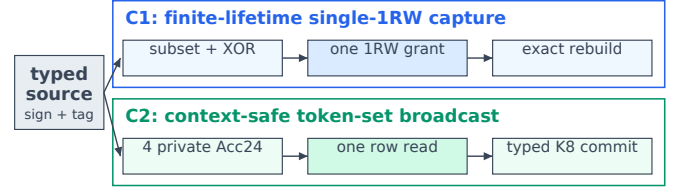


Fig. 1. Two exact reuse objects under one typed-source protocol. C1 retains selected parents in a finite 1RW store; C2/TSBG broadcasts one delivered weight row while preserving private signed products and Acc24 contexts.

the same weights. C1 retains selected parents for the current tile, schedules reads and writes through one physical port, and suppresses writes without a future consumer. C2 shares an Acc24 fabric across eight typed sources; its TSBG frontend groups consumers before weight reads so that one delivered row serves several contexts. Signs, destinations, products, and accumulators remain private. The common design principle is to decide reuse before incurring the access or computation it eliminates. The frozen Motion checkpoint supplies the workload; neither circuit requires retraining or changes to its source descriptors.

## I. WORKLOAD AND ARITHMETIC MODEL

The frozen Motion C12 ep34 checkpoint obtains 1.199514 average endpoint error (AEE) on an 825-frame local DSEC validation set and has a 5.6709% global firing rate. Hardware evaluation consumes the sealed ep34 source descriptors generated by fixed normal inference. Captured ATLIF sources entering the evaluated operators are binary; signed source codes represent downstream polarity and correction traffic rather than analog tensor values. The frozen software path uses Motion-XOR scoring with  $\alpha=0.125$  and K as its value carrier, while hardware quantization is disabled on that path. Model, VCS, and physical evidence remain explicitly tagged and are never multiplied.

Bottleneck convolution exposes repeated products only while parent and destination state remain resident. That invariant defines C1. Attention and decoder traces remain part of the frozen identity but are not promoted to performance contributions.

## II. C1 MICROARCHITECTURE

### A. Directory and Legal Parents

C1 first builds a ping-pong directory for each 64-row, 16-source mask task. For a row with at least two active

sources, C1 selects the nonempty exact-subset parent with maximum population count, breaking ties by original row index. Equal-mask parents must have an earlier original index. Execution follows stable population-count/index order. Both the one-parent forest and this dependency-respecting order are inherited from the Prosperity artifact [1]; they are not our algorithmic contribution. A static parent-needed bitmap marks rows selected by at least one child. A row with no selected child is never written to parent scratch. This is unused-result write suppression, not dynamic last-consumer reclamation.

The 240-KiB-class budget covers the modeled working set, not parent scratch alone. Nine  $128 \times 128$  1RW SRAM leaves operate in parallel as one  $128 \times 1152$  parent-product scratch and retain 96 signed 12-bit product lanes. Masks and the live-parent directory are control state; the downstream psum store is a separate boundary. The current 64-row task addresses only 64 entries: selected parents are retained until task completion, without within-task eviction or allocation-pressure misses. The engine never assumes the ideal simultaneous 1R1W access that produced a higher but unphysicalized reuse ceiling. The circuit contribution is implementing this established reuse relation with single-port service, write suppression, and atomic completion; the measured ratio against zero skipping also includes the inherited product-sparsity benefit.

### B. Residual Issue and Exact Reconstruct

A selected parent mask  $\mathcal{M}_p \subset \mathcal{M}_r$  yields a disjoint residual. C1 issues only that residual and reconstructs the 96-lane signed product row by adding the stored parent:

$$\mathbf{P}_r = \mathbf{P}_p + \sum_{s \in \mathcal{M}_r \setminus \mathcal{M}_p} \mathbf{p}_s. \quad (1)$$

Exactness follows from the disjoint union, not from value prediction. If no legal nonempty parent exists during directory construction, C1 selects the empty parent and issues the original source row. Sign, destination, and accumulator identity are never coalesced across rows. Thus sparsity changes delivered work without changing the operator result.

### C. Single-Port Arbiter and Atomic Commit

The execution side combines one earliest-parent lookahead, a two-entry reserved response queue, and forwarding under a deadline-aware 1RW arbiter. A parent-scratch read and a retained-result write compete for the same macro port; residual arithmetic can overlap a read, and architectural psum commit is a separate interface. The arbiter grants the access that would otherwise miss its row-completion deadline; the reserved queue absorbs one already-accepted response so that a later grant cannot orphan it. Forwarding supplies a just-completed parent to a waiting residual without a macro read.

Final psum write and row completion are a single atomic event. Backpressure may change service order, but it cannot expose a partially reconstructed row: the typed-source terminal bit and the completion pulse retire together. A directed VCS tile on the first 64 real support masks of the cycle ledger is used later only to calibrate these event counts, not to replace the same-ledger cycle model.

Exactness depends on the subset relation and producer-before-consumer ordering. The written bitmap distinguishes scratch-resident results from not-yet-written producers; forwarding is an alternative source of a valid parent. Protocol checks do not implement eviction recovery. Alternative parents, dynamic reclamation, and lifetime-minimizing row schedules are not part of this mapped island.

### III. C2 TYPED-K8 AND CONTEXT-SAFE BROADCAST

The C2 fabric accepts up to eight nonzero typed sources and shares the Acc24 array, endpoint, and control rather than replicating eight K1 engines. Its fair equal-service baseline is therefore  $K1 \times 8$ , not one K1 lane. Eight independent memory-bank channels may return out of order; epoch, slot, generation, tag, slice, and source-channel fields bind every response before it can enter the signed bridge. A stale or misrouted response is rejected rather than accumulated.

TSBG changes only weight delivery. Four token contexts expose a common source-group row, so one fetched row can serve four independent signed products and four Acc24 contexts. For shared row  $W_g$ ,

$$\text{fetch}(W_g) = 1, \quad A_c \leftarrow A_c + v_c W_g, \quad (v_c, d_c, A_c) \text{ private}. \quad (2)$$

It is incorrect to share the product because equal source indices can carry different signs or destinations. Both ordinary and broadcast modes elaborate the same 192-bit row-live bitmap, four-row cache, public memory ports, and hierarchical priority encoder; only the static scheduling order changes from token-major to source-group-major. A cache hit is resolved before the bank request state, so a reused row suppresses the memory request and all downstream response traffic instead of merely gating the multiplier after a read.

Exact fallbacks are explicit. If token contexts disagree on a weight-row identity, C2 emits separate fetches. If C1 finds no legal nonempty parent when building its directory, it issues the original source row. Backpressure may alter service order but not ownership: terminal and atomic commit retire only after the corresponding private accumulator is complete.

Figure 2 makes the saved circuit event explicit. Ordinary token-major execution resolves the same row separately for each context. TSBG first proves a common row identity, then asserts one SRAM request and fans out the returned row. The other three read-enables, addresses, and responses are never generated; this is memory-traffic suppression, not a multiplier gate applied after the access. Product formation and retirement remain four private operations.

### IV. EVALUATION

All physical results use a commercial 28-nm flow at 3.0 ns. Reported points are prelayout, use ideal clocks and ZeroWireload, and have no extracted parasitics. C1 area includes nine generated SRAM leaves; C2 comparisons are logic-only and therefore remain separately labeled in Table IV.

Table I exposes the baseline ladder instead of attributing all inactivity to product capture. The ledger is 51.84 million source rows from ten zurich\_city\_09\_a ep34 samples. Per-sample ratios span  $1.661\text{--}1.723\times$ , with a  $1.6946\times$  geometric

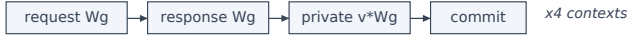
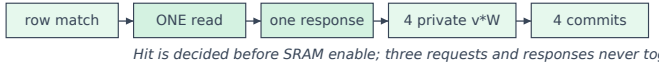
**Ordinary token-major delivery****TSBG source-group-major delivery**

Fig. 2. TSBG suppresses redundant SRAM transactions before request. The returned row fans out to private signed products and Acc24 contexts; component ratios from different evidence classes are not multiplied.

TABLE I  
C1 SAME-LEDGER CAPTURE LADDER (51.84 M SOURCE ROWS).

| Execution point     | Cycles (M) | vs. zero | Boundary     |
|---------------------|------------|----------|--------------|
| Strong zero         | 648.741    | 1.000×   | baseline     |
| Same-coordinate bit | 646.619    | 1.003×   | baseline     |
| C1 finite 1RW       | 382.849    | 1.6945×  | cycle model  |
| Concurrent-access   | 341.058    | 1.902×   | ceiling only |

mean. Strongest-zero skipping still walks parent-state traffic for every nonzero source; same-coordinate bit skipping on this ledger removes almost none of that traffic (1.003×). Exact parent reuse is what creates the material reduction. The remaining 12.25% gap from C1 to the concurrent-access ceiling is the tax of the physical single-port contract. We do not promote the ceiling as an attainable RTL result.

The cycle model charges one 1RW grant per accepted macro access, residual issue, forwarding, deadline holds, and atomic commits. Its cycles are modeled service events, not CPU runtime or a full-ledger RTL replay.

The mapped nine-SRAM island occupies 166,514  $\mu\text{m}^2$ . Mapped synthesis and prelayout PrimeTime use the foundry 28-nm standard-cell library and the nine compiler SRAM abstracts; no Yosys or educational library is used as an ASIC PPA proxy. PrimeTime reports setup/hold slack of +27.9/+1.8 ps at 3 ns under ideal clocks. Formality passes 16,549 mapped-to-mapped compare points on that netlist pair. In a separately bounded 64-row, 253-cycle directed window the island dissipates 29.08 mW and 22.07 nJ. That energy is a mixed-corner, prelayout PrimeTime PX estimate without SPEF and without a frame-level activity factor; it is not energy per DSEC frame and is not a chip measurement.

A separately sealed ep34 tile uses the first 64 real support masks of the cycle ledger. VCS matches 196 accepted issues, 58 parent edges, 31 dead-write elisions, 54/33 macro reads/writes, four forwards, six deadline holds, 14 stalls, and 64 commits/completions. Lane values are deterministic synthetic signed-12-bit data with zero prior psum. This calibrates modeled event counts and directed arithmetic; it does not itself produce the 1.6945× cycle ratio. A 64-row task is the scheduling quantum, not the convolution depth; no parent replacement occurs within it.

For numerical binding, eight retained Conv3x3/ConvTranspose weights use per-output-channel narrow-range INT8 with a power-of-two scale. The largest absolute-code bound is 200,219, below Acc24; 1,280 Python-integer probes match and the observed final range is [−29,680, 27,619]. On 825 local DSEC frames the hardware-

TABLE II  
TSBG-B4 FIXED 40-SAMPLE VCS SCOPE LADDER.

| Scope                 | Loads        | Ordinary        | TSBG            | Speedup        |
|-----------------------|--------------|-----------------|-----------------|----------------|
| G48: FC1+4 FC2        | 1,920        | 12.523 M        | 5.124 M         | 2.4438×        |
| G96/192: 8 FC2        | 960          | 80.129 M        | 45.381 M        | 1.7657×        |
| <b>all identities</b> | <b>2,880</b> | <b>92.652 M</b> | <b>50.505 M</b> | <b>1.8345×</b> |

order mixed-precision candidate obtains 1.197367 AEE versus a 1.199514 historical baseline, passing the predefined +0.02 gate. Different GPU backend flags make this a compatibility check, not a causal accuracy improvement.

TSBG is evaluated on 2,880 fixed ep34 workloads selected independently of performance: 40 samples from four DSEC sequences, all 12 FC1 and 12 FC2 layer identities, and first/middle/last aligned B4 token quartets. The G48 engine handles all FC1 and four FC2 identities directly; the remaining six G96 and two G192 FC2 identities execute as functionally exact 48-source continuations. Activity comes from the frozen checkpoint, whereas arithmetic uses deterministic directed INT8 verification weights rather than captured model weights. Naturally nonzero descriptors in this population carry +1; directed negative-source tests separately cover the signed protocol. The G48 population covers 1,920 workloads and the continuation batch covers 960. Both use identical public ports and cache capacity; Table IV reports post-load execution. The G48 subset spans 2.381–2.546× by sequence. The continuation population contains 554 improvements, 404 ties, and two regressions of one and seven cycles (worst 0.99987×); these tails remain in the ratio of sums.

The shared verification memory uses bank-dependent one-to-nine-cycle responses, periodic request backpressure, and independent issue/commit stalls. These service assumptions are identical across modes, not measured foundry-SRAM latency. Consequently, the ratios characterize this frontend and its response-assembly contract; they do not establish superiority over an arbitrary fully overlapped streaming architecture. A faster memory or another refill/compute overlap policy requires a new matched comparison, rather than multiplying a separate streaming gain into Table II.

To test whether the three fixed B4 regions hide a position effect, we calibrate an independent cycle recurrence against all 2,880 VCS workloads and then replay every aligned B4 quartet in the same frozen 40-sample capture. Table III reports this model separately from RTL execution. Across 11.16 million quartets, TSBG reduces modeled cycles from 313.604 to 150.234 billion (2.0874× ratio of sums, or 52.09%) and scheduled scalar weight-read requests from 192.483 to 67.992 billion (64.68%). The four sequence ratios span only 2.0807–2.0968×; FC1 and FC2 obtain 2.1766× and 1.9005×, respectively. Per-quartet p10/p50/p90 are 1.0000/1.5414/2.4388×. Because the mode is elaboration-time, there is no dynamic fallback: 3.10% of quartets are marginally slower and the worst case is 0.99755×.

For  $G \leq 48$ , the recurrence matches 3,840 ordinary/TSBG VCS cycle fields with zero fitted residual. G96/G192 comprises 780,000 quartets, or 6.981% of the full population; 779,040 (99.877% within this stratum) use the median wrapper

TABLE III  
ALL-ALIGNED-B4 ROBUSTNESS [VCS-CALIBRATED CPU MODEL].

| Scope                      | Ordinary cycles | TSBG cycles | Ratio            |
|----------------------------|-----------------|-------------|------------------|
| All 24 FC layers           | 313.604 G       | 150.234 G   | 2.0874×          |
| FC1 / FC2                  | —               | —           | 2.1766 / 1.9005× |
| Four sequences             | —               | —           | 2.0807–2.0968×   |
| Observed-residual envelope | —               | —           | 2.0870×          |

TABLE IV  
ADMITTED COMPONENT EVIDENCE; OPEN CLAIMS ARE NOT INFERRED.

| Class    | Result                                                                                 | Boundary                                                                               |
|----------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| C1       | 1.6945×; −40.99% time vs. strong zero <sup>[model]</sup>                               | Ten-sample same-ledger model; not RTL/system speedup                                   |
| K8       | 1,913/1,945 cycles; 4.541× throughput/logic-area; logic −77.61% <sup>[VCS][DC]</sup>   | vs. equal-bandwidth K1×8; five directed loads; logic-only                              |
| TSBG     | 92.652/50.505 M cycles (1.8345×); reads −58.13% <sup>[VCS]</sup>                       | 2,880 real-activity loads; all 12 FC1/FC2 identities; fixed three regions, not full FC |
| C1 phys. | 166,514 $\mu\text{m}^2$ ; +27.9/+1.8 ps; 29.08 mW / 22.07 nJ <sup>[DC/PT]</sup>        | Nine-SRAM prelayout island; mixed-corner directed window; no SPEF                      |
| C1 store | 18,432 B / 9 macros; 214,912 B / 105 macros / 0.988 mm <sup>2</sup> <sup>[model]</sup> | Parent scratch integrated above; complete ledger is an area model, not integrated PPA  |

residual from 960 keyed VCS anchors. Replacing all such ordinary/TSBG residuals by their observed minimum/maximum changes the aggregate only from 2.087430× to 2.086979×. This is an observed-envelope sensitivity, not a formal bound. Inputs retain real ep34 activity/sign descriptors, while arithmetic weights remain directed timing values; the result is a VCS-calibrated FC component model, not RTL, same-area, or whole-network execution.

*Pre-read causality.* A separate directed VCS ablation compares ordinary token-major execution, group-major post-read gating, and group-major pre-read TSBG. The post-read control performs the same scheduling but still requests, receives, and identity-checks the redundant weights before discarding them. The three axes issue 2,304/2,304/576 scalar reads, respectively, while each completes 4,608 signed products and 24 commits with zero mismatch. Thus the 75% request reduction in this directed case occurs before SRAM access; reordering followed by post-read gating does not produce it. This isolates request causality, not matched area or energy, and is separate from the 2,880-workload result.

The matched clock-gated ordinary/TSBG implementations occupy 234,537/235,701  $\mu\text{m}^2$  (+0.50%). Common 3-ns constraints, 200-ps setup uncertainty, and 50-ps hold uncertainty are preserved through clock gating and hold-buffer repair. Setup slack is +122.7/+72.4 ps; reported hold slack is positive but below 0.01 ps, not a robust post-CTS margin. Mapped-to-mapped Formality passes 77,180/77,155 compare points. The earlier ungated pair was +0.0118% in area with hold open; that ratio does not describe the gated implementation. The common 288-KiB, 16-macro weight capacity is a 558,507- $\mu\text{m}^2$  foundry-QRT area model and is not counted as logic or as capacity saved by TSBG. The observed 58.13% reduction is bank activation, not deployed SRAM area. For the equal-bandwidth K8 comparison, adding that identical common capacity to both axes changes the K8/K1×8 area from 131,086/585,479

TABLE V  
MATCHED CLOCK-GATED POST-LOAD LOGIC ENERGY [PRELAYOUT PTPX].

| Reuse stratum | Ordinary / B4 cycles | Energy (nJ), O / B4 | Reduction |
|---------------|----------------------|---------------------|-----------|
| Low           | 4,717 / 4,717        | 310.67 / 346.48     | −11.53%   |
| Median        | 6,733 / 4,465        | 443.10 / 327.25     | 26.15%    |
| High          | 22,294 / 7,554       | 1,467.03 / 555.64   | 62.12%    |

to 689,593/1,143,986  $\mu\text{m}^2$ . The resulting foundry-QRT logic-plus-memory model retains a 39.72% area reduction and 1.687× directed-throughput/area advantage. It is an arithmetic model over matched capacity, not an integrated macro placement or routed timing result.

*Reuse density and execution energy.* Table V uses three windows selected from low/median/high reuse-density strata before power measurement. PrimeTime PX consumes post-load, zero-delay gate-VCS activity at TT, 0.9 V, 25 °C with 0.1-ns ideal-clock slew. Both netlists use common automatic clock-gating settings. All 1,712 inputs and 74,349/74,324 register outputs receive direct gate activity. Weights are checkpoint-derived candidate INT8 FC weights, not an admitted FC quantization accuracy result. SRAM bank energy, preload, CTS, routed glitches, and parasitics are excluded. B4 averages 24.43–24.52 mW versus 21.93–21.95 mW for ordinary: different clock-enable groupings leave B4 with higher clock-pin power. Low reuse therefore *increases* logic energy by 11.53%; medium/high reuse saves energy through shorter execution, not lower power per cycle. These are density sensitivities, not frame averages.

## V. RELATED WORK

Prosperity already discovers subset/prefix parents, issues residual activations, and dependency-schedules their reconstruction [1]; C1 does not reclaim those ideas. Its circuits claim is their execution under a tile-resident parent-product store and one physical 1RW port, including deadline arbitration, reserved response, forwarding, unused-parent write suppression, and atomic completion. C1 inherits stable population-count ordering; it does not actively shorten parent lifetime. Phi stores hierarchical patterns chosen by DSE [2]. We adopt their evaluation pattern—cycle simulation plus RTL/physical anchors and a baseline ladder—without relabeling their numbers as ours.

FireFly-T already broadcasts wider vectorized data from one memory bank across spatial-temporal work [8]; ELSA, SpikeX, and Bishop establish bundled delivery and spatial/temporal weight reuse [9], [10], [11]. TSBG does not reclaim broadcast, bundling, group-major order, or weight reuse. Its specialization decides a row hit before the SRAM request and shares only that delivery among typed signed contexts while destination, tag, terminal, product, and Acc24 ownership remain private under the same cache/port contract. This direct prior also precludes comparing K8 against a single K1 lane.

SpinalFlow and SNE establish event-driven execution and state locality [4], [5]; ODIN demonstrates a silicon neuromorphic core [3]. Their PPA is not an interchangeable baseline for these digital reuse islands. TCAS-II WS-LOS combines weight/local-output reuse with skipping [12], while delta

TABLE VI

DIRECT-PRIOR BOUNDARY; INHERITED MECHANISMS ARE NOT RECLAIMED.

| Prior                | Inherited mechanism                         | Circuit specialization here                                   |
|----------------------|---------------------------------------------|---------------------------------------------------------------|
| Prosperity           | subset parent, residual, dependency         | tile-resident 1RW service; atomic completion                  |
| FireFly-T / ELSA     | bank broadcast; bundled Gustavson delivery  | typed sign/destination/Acc24 private; hit before read request |
| Phi                  | hierarchical sparse pattern and DSE         | no pattern claim; physical port/capacity ladder               |
| WS-LOS               | weight/local-output reuse and skipping      | same-cache token-set A/B with private products                |
| Delta weight sharing | static representation and traffic reduction | unchanged weights; runtime pre-read context broadcast         |

weight sharing reduces traffic through static inter-network representation [13]. Our weights are unchanged; the evaluated event is a runtime read shared by private contexts. Request counts, logic cost, and memory energy therefore remain separate metrics.

A 28-nm optical-flow chip separates operations, memory traffic, latency, energy, and AEE [14]. We likewise separate same-workload component comparisons, public-artifact opportunities, and published chip results, without ranking prelayout estimates against silicon energy.

## VI. SCOPE AND LIMITATIONS

C1 reuses products under a 1RW service constraint; C2/TSBG reuses weight delivery while retaining private accumulation. Their ratios are not multiplied, and neither proves a whole-network latency improvement.

In the 2,880-workload direct-VCS population, TSBG has seven slightly slower nonempty cases because group-major order can lose ordinary-cache residency under weak reuse. These cases remain in the ratio of sums. The reported population spans every FC1/FC2 layer identity but uses three fixed B4 regions per layer/sample; it does not imply full-FC wall time. Ordinary scheduling remains an exact fallback.

C1's mapped nine-SRAM island and full 214,912-B ledger are separate evidence; the 105-macro capacity is a 0.988-mm<sup>2</sup> area model rather than integrated PPA. Its 22.07-nJ result is a directed prelayout window, not frame energy. C2/TSBG has prelayout gated setup/hold repair and a three-window logic-energy comparison; post-CTS timing robustness and SRAM-inclusive power remain open. Its common 288-KiB weight store is modeled separately. No component ratio is promoted to monolithic FPS, energy/frame, or whole-network speedup.

## VII. CONCLUSION

Exact reuse requires matching data lifetime to physical service. C1 combines finite parent residency with single-port reconstruction; C2/TSBG combines shared weight delivery with private signed accumulation. Their evaluation separates modeled product-capture latency, RTL-observed read suppression, and 28-nm circuit cost. The resulting component design targets redundant transfers that remain after ordinary zero-source skipping, without changing the evaluated source arithmetic.

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